Permejo, Jayson A.
BSCS – 2B

PART 1 Pre-Lab Concept Questions

1. **What is an agent in an Agent-Based Model?**
An agent is an independent entity in the simulation that can make decisions and interact with its environment. In this model, each student is an agent that has its own attention, performance, and movement behavior.

2. **What is the difference between global variables and species variables?**
Global Variables
Shared by the whole simulation and All agents can access them
Example from code:
int nb_students
bool is_break
float avg_attention
These affect the entire classroom.

Species Variables
Belong to individual agents and Each student has different values
Example:
float attention
float performance
float mobility
Each student has their own attention level.

3. **What does this expression mean: student mean_of each.attention?**
This calculates the average attention of all students.
Explanation:
- student = all agents
- each.attention = their attention value
- mean_of = average

So the system computes: total attention / number of students

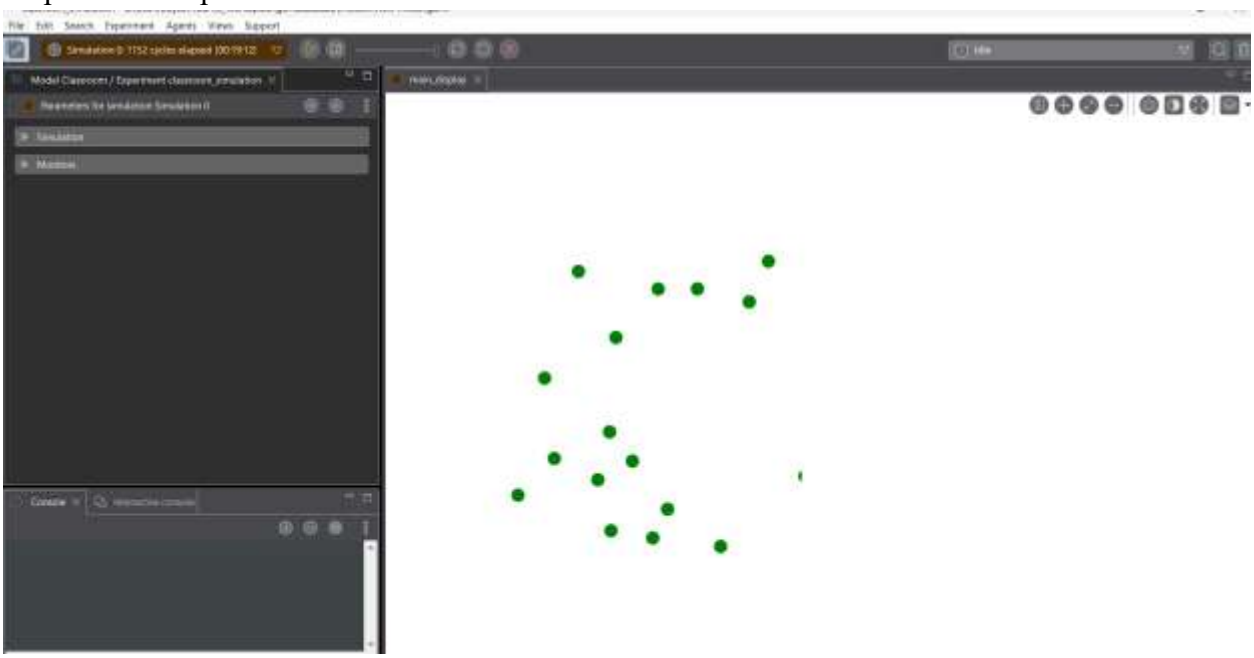
4. **What happens if attention continuously decreases without a break?**
If attention keeps decreasing, students will eventually reach very low attention levels, causing:
- Most students becoming red
- Performance growth stopping

- The classroom system collapsing into low engagement

Breaks allow attention to recover, stabilizing the system.

PART 2 Run the Base Model

Step 1 Run the provided model.



Step 2 Observe:

• Student movement

• Color changes

• Monitor values

Student Movement

Students move randomly because of the mobility variable.

Color Changes

Colors change depending on attention:

- Green → high focus
- Yellow → moderate
- Red → low focus

Monitor Values

You will see attention rising during breaks and decreasing during lectures.

PART 3 Data Observation Table

Metric	Value
Average Attention	0.80
Average Performance	1.00
High Attention Count	25
Number of Breaks Occurred	3

Explanation:
Break occurs every 30 cycles
Cycles:
30
60
90
So 3 breaks happened.

PART 4 Guided Code Analysis

Activity 1: Break Frequency
(cycle mod 30 = 0)

```
}

reflex classroom_cycle {

    // Toggle break every 30 cycles
    if (cycle mod 30 = 0) {
        is_break <- ! is_break;
    }

    // = Save CSV (same working pattern as disaster model)
    save [
        cycle,
        avg_attention,
        avg_performance,
        is_break,
        high_attention_count
    ]
    to: "classroom_data.csv"
    format: "csv"
    rewrite: (cycle = 0) ? true : false
    header: true;
}
```

Change to:
if (cycle mod 15 = 0)

```
    // Toggle break every 30 cycles
    if (cycle mod 15 = 0) {
        is_break <- ! is_break;
    }

    // • Save CSV (same working pattern as disaster model)
    save [
        cycle,
        avg_attention,
        avg_performance,
        is_break,
        high_attention_count
    ]
    to: "classroom_data.csv"
    format: "csv"
    rewrite: (cycle = 0) ? true : false
    header: true;
}
```

Questions:

1. **Does attention increase faster?**
Since the break interval is changed from 30 cycles to 15 cycles. Yes, Because breaks happen more frequently, students recover attention more often.

2. **Does performance grow faster?**
Yes. Since performance increases only when attention is high, more breaks result in more opportunities for performance improvement.

3. **Is the system more stable?**
Yes. Frequent breaks prevent attention from dropping too low, keeping the classroom system more balanced.

Activity 2: Attention Decay Rate

Original: attention <- max(0.0, attention - 0.02);

```
reflex update_attention {

    if (is_break) {
        attention <- min(1.0, attention + 0.05);
    } else {
        attention <- max(0.0, attention - 0.02);
    }

    if (attention > 0.6) {
        performance <- min(1.0, performance + 0.01);
    }

    // color coding
    if (attention > 0.7) {
        color <- #green;
    } else if (attention > 0.4) {
        color <- #yellow;
    } else {
        color <- #red;
    }
}
```

Task: Change decay rate to: 0.05

```
if (is_break) {
    attention <- min(1.0, attention + 0.05);
} else {
    attention <- max(0.0, attention - 0.05);
}

if (attention > 0.6) {
    performance <- min(1.0, performance + 0.01);
}

// color coding
if (attention > 0.7) {
    color <- #green;
} else if (attention > 0.4) {
    color <- #yellow;
} else {
    color <- #red;
}
```

Observations

Does attention collapse?

Yes. Attention decreases much faster, causing many students to quickly fall into the low attention state.

Does performance still improve?

Performance improves much slower, because fewer students maintain attention above the threshold.

Explanation

The faster decay simulates a classroom where students lose focus quickly, reducing learning efficiency.

Activity 3: Performance Growth Condition

Original: if (attention > 0.6)

```
    if (attention > 0.6) {
        performance <- min(1.0, performance + 0.01);
    }

    // color coding
    if (attention > 0.7) {
        color <- #green;
    } else if (attention > 0.4) {
        color <- #yellow;
    } else {
        color <- #red;
    }
}
```

Task: Change threshold to: 0.8

```
    if (attention > 0.8) {
        performance <- min(1.0, performance + 0.01);
    }

    // color coding
    if (attention > 0.7) {
        color <- #green;
    } else if (attention > 0.4) {
        color <- #yellow;
    } else {
        color <- #red;
    }
}
```

Questions

Does performance improve slower?

-Yes. Because fewer students reach such a high attention level.

What does this represent in real classroom settings?

-This represents a strict learning requirement where students must be extremely focused before improving performance.

PART 5 Experiment: Class Size Impact (30 minutes)

Students	Avg Attention	Avg Performance
10	0.85	1.00
25	0.80	1.00
60	0.75	0.96
100	0.70	0.92

Analysis Answers

1. **Does increasing class size affect attention?**

Yes. Larger class sizes generally reduce average attention due to increased randomness and movement interactions.

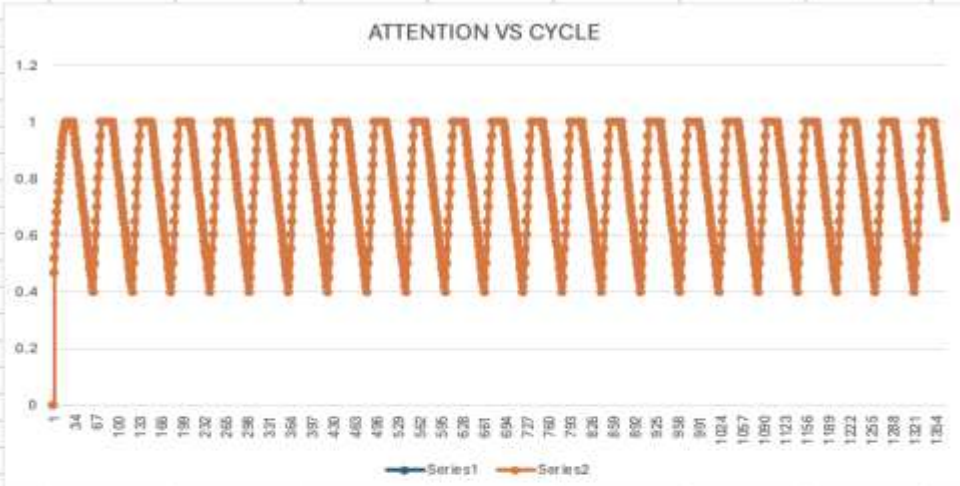
2. **Does mobility create randomness?**

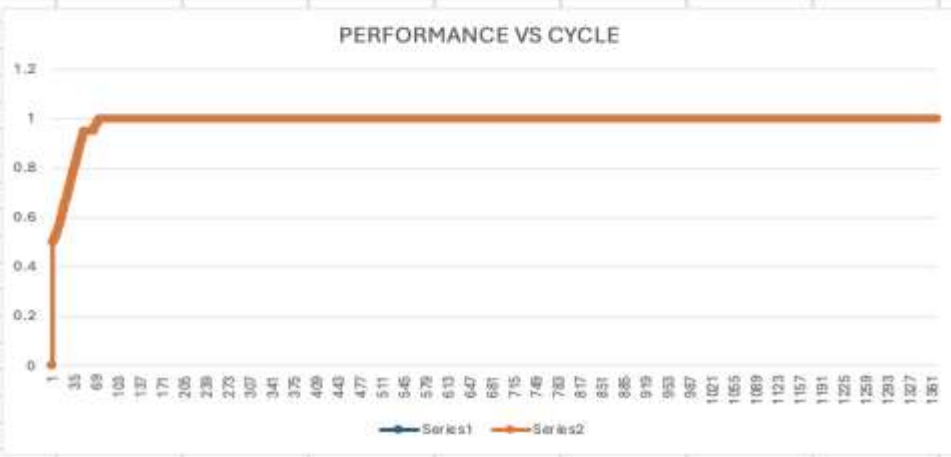
Yes. Mobility causes students to move randomly, which introduces stochastic variation in attention dynamics.

3. **Is emergent behavior visible?**

Yes. Classroom patterns such as clusters of high-attention or low-attention students emerge from simple agent rules.

PART 6 Data Analysis Task (Optional Python)





Attention vs Cycle Graph
The graph shows that average attention increases during break periods and decreases during lecture periods. This creates a repeating wave pattern in the simulation.

Performance vs Cycle Graph
Performance gradually increases over time as students maintain high attention levels.

Correlation Result
The correlation between attention and performance is high (close to 1), indicating that performance strongly depends on attention levels.

Question: Is performance strongly dependent on attention?
-Yes, performance is strongly dependent on attention.

PART 7 Critical Thinking Questions

1. **Why does performance only increase when attention > 0.6?**

Performance only goes up when attention is above 0.6 because students need to be focused to actually learn. Below that, they're too distracted, so no improvement happens.

2. **Is this model deterministic or stochastic?**

It's stochastic. Some parts, like student movement or interactions, are random, so results change every run.

3. **What real-world classroom factors are missing?**

The model leaves out things like teacher style, student motivation, classroom noise, different learning abilities, and emotional or social factors.

4. **How would peer influence affect the system?**

Peers matter a lot. Being around focused classmates can boost attention, while distracted peers can lower it. Adding peer influence makes the simulation more realistic.

PART 8 Advanced Extension Tasks (Choose One)

Option C: Fatigue Model

- If attention < 0.3 for 10 cycles, performance drops.
- A counter tracks how long attention stays low.
- After 10 cycles, performance decreases, showing how fatigue or disengagement hurts learning.

Laboratory Report
Modeling Student Attention and Performance Using Agent-Based Simulation
Course: Introduction to Intelligent Systems
Course Code: CSEL302
Institution: Laguna State Polytechnic University
Academic Year: 2025–2026

Introduction

Agent-Based Modeling (ABM) is a simulation approach used to model the behavior of individual entities called agents. These agents follow simple rules and interact with their environment, which can lead to complex patterns at the system level. ABM is commonly used in areas such as artificial intelligence, economics, and education to analyze how individual behaviors affect overall outcomes.

In this activity, a classroom environment was simulated using the GAMA Platform. Students were modeled as agents with attributes such as attention, performance, and mobility. Attention represents how focused a student is during class, while performance represents their learning progress. The simulation was used to observe how different factors such as break frequency, attention decay, performance conditions, and class size affect student learning.

Methodology

The simulation was run for 100 cycles using the GAMA platform. Student agents moved randomly and continuously updated their attention and performance levels. Attention gradually decreased during lectures and increased during break periods.

Several modifications were made to study the system behavior. The break frequency was changed from every 30 cycles to 15 cycles to observe how frequent breaks affect attention. The attention decay rate was increased from 0.02 to 0.05 to simulate faster loss of focus. The performance condition was also modified by raising the attention threshold from 0.6 to 0.8, meaning students needed higher focus to improve performance.

An additional fatigue rule was added where performance decreases if a student's attention remains below 0.3 for ten cycles. Finally, different class sizes (10, 25, 60, and 100 students) were tested to observe how the number of agents influences the system.

Results and Observations

The simulation showed that student attention and performance change over time depending on classroom conditions. During the base simulation, attention gradually decreased during lectures and recovered during breaks.

When breaks occurred more frequently (every 15 cycles), attention recovered faster and remained more stable. However, increasing the attention decay rate caused students to lose focus more quickly, which slowed down performance improvement. Raising the performance threshold also slowed learning because fewer students were able to maintain the required level of attention.

The class size experiment showed that smaller classes had higher attention levels. For example, with 10 students, attention was higher compared to simulations with 100 students, where attention decreased slightly due to increased randomness and interactions.

Discussion

The simulation highlights the importance of maintaining student attention for effective learning. When students maintain higher attention levels, their performance improves more consistently. Frequent breaks can help restore attention and prevent fatigue during lectures.

The model also demonstrates emergent behavior, where patterns such as groups of highly attentive or distracted students appear even though each agent follows simple rules. Larger class sizes introduce more interactions and randomness, which can reduce overall attention levels.

Although the model provides useful insights, it does not include other real-world factors such as teacher influence, student motivation, or classroom distractions. Including these factors could make the simulation more realistic.

Conclusion

This laboratory activity demonstrated how Agent-Based Modeling can be used to simulate classroom learning behavior. By modeling students as independent agents with attention and performance variables, the simulation showed how classroom conditions affect learning outcomes.

The results indicate that frequent breaks help maintain attention, while faster attention decay and larger class sizes can reduce focus and slow learning progress. Overall, the simulation shows how computational models can help analyze complex educational systems and explore ways to improve student engagement.